BASE CHANGE ALONG AN EXTENSION OF A SEPARABLY CLOSED FIELD INDUCES A BIJECTION ON IRREDUCIBLE COMPONENTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written; the author's sentence 'Since X' = Z times_k X the result is a special case of Lemma ...' is reproduced verbatim and the cited lemma is carried as a navigation stub.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-separably-closed-field-irreducible-components is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- An extension of a separably closed field is geometrically irreducible. Algebra, Lemma algebra-lemma-separably-closed-irreducible-implies-geometric, invoked at proof lines 1532-1534. Supplies geometric irreducibility of k' over k.
- Local criterion for geometric irreducibility of a scheme. Same chapter, Lemma lemma-geometrically-irreducible-local, invoked at proof lines 1535-1536. Supplies that the spectrum of k' is geometrically irreducible over k.
- Projection from a geometrically irreducible factor induces a bijection on irreducible components. Same chapter, Lemma lemma-bijection-irreducible-components, invoked at proof lines 1537-1538. Supplies the conclusion once the base change is seen as such a product.

Source reference. Original lemma 037M: sources/stacks-chow/tags.txt, source lines 4205–4205. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038F: sources/stacks-varieties/pinned-original-source.tex, source lines 1380–1380. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038G: sources/stacks-varieties/pinned-original-source.tex, source lines 1405–1405. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k'/k be an extension of fields. Let X be a scheme over k. Set $X' = X_{k'}$. Assume k separably algebraically closed. Then the morphism $X' \to X$ induces a bijection of irreducible components.*

Proof. Since k is separably algebraically closed we see that k' is geometrically irreducible over k, see Algebra, Lemma 037M. Hence $Z = \mathrm{Spec}(k')$ is geometrically irreducible over k. by Lemma 038G above. Since $X' = Z \times_k X$ the result is a special case of Lemma 038F. □